MOE
Innovation
Programme

Introduction to the IvP

Explore the World of Design



About MOE Innovation Programme (IvP)

The Innovation Programme (IvP), which started in 1993, provides a unique opportunity for **Primary 5 and Secondary 2/3 students** to:

- Develop inventive thinking skills and dispositions,
- Be exposed to knowledge and experiences beyond the classroom through interaction with innovation mentors from tertiary institutions and other organisations

Timeline



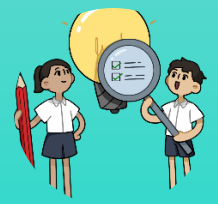
IvP School-based Launch

During 1st IvP Lesson



3 Draft Sessions with Innovation Mentors

Week of 8 Apr
Week of 6 May
Week of 8 Jul



School-based Assessment

End-Term 3



Young Innovators' Fair

7 Nov (PM)

Partners

Nanyang Technological University, National University of Singapore, Singapore University of Technology and Design, Nanyang Polytechnic, Ngee Ann Polytechnic, Republic Polytechnic, Singapore Polytechnic, Temasek Polytechnic, Innovators and Entrepreneurs Association (IdEA), and 3M Singapore

Explore the World of Design

In this lesson, we will dive into the fascinating world of design. Together, we will explore what design truly means and uncover the vital role it plays in shaping the world around us.

What is design, and how is it different from art?

What are the key elements that make good design?

How does design impact our daily lives?



Be a Design Detective!



Explore examples of everyday designs to uncover their purposes, understand their intentions, and evaluate if they are beneficial for users.





What makes a good clock?



How do we use this?







Which is a better design? Why?









How is this an improvement over existing solutions?





Watch a short segment of this video to understand how the Scrub Daddy works. If you'd like to gain further inspiration from the innovator's journey, feel free to watch the entire clip.







Does this design achieve its purpose? Why or why not?





What have you learnt about design?

1. What is design?
2. What makes good design?



Design is about solving problems and helping people, not just making things look good.



Does this design achieve its purpose? Why or why not?

Good design is about CARE.

- C - Care for the User
- A - Achieve its Goal
- R - Respect Broader Impact
- E - Enhance lives



C - Care for the User

Does the design meet the needs, values, and preferences of the user?







A – Achieve Its Goal

Does the design solve the problem well and improve on what's already available?



Which design is better?





R – Respect Broader Impact

Does the design consider its broader effects on society, the environment, and long-term sustainability?

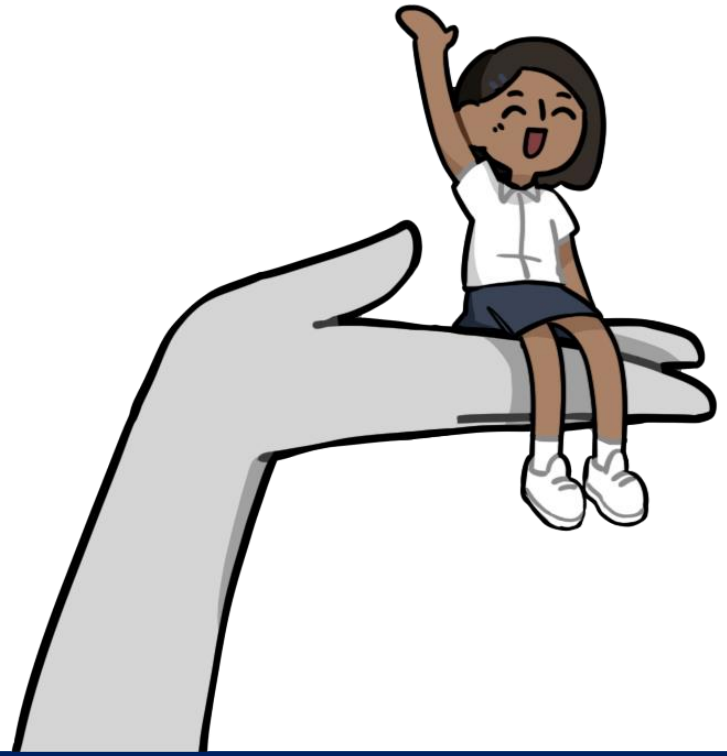




REUSABLE – Not Your Average Cup

E - Enhance lives

Does the design improve the user's quality of life, making it better, easier, or more enjoyable?





Flow Wall Desk

Is Sitpack good design?

Do you see CARE in this design?





Sitpack: The Compact Foldable Seat That Fits In Your Pocket

C - Care for the User

The Sitpack is thoughtfully designed to meet the needs of users who require a compact, portable seating solution. Its lightweight and foldable design makes it easy to carry around, while the telescopic feature allows for quick setup and adjustment to the user's height, ensuring ergonomic comfort during use.







A – Achieve Its Goal

Compared to other portable seating options like folding chairs or cushions, the Sitpack offers greater convenience due to its small size, ease of use, and comfortable sitting position. Its telescopic design makes it a more compact solution than larger, bulkier alternatives.

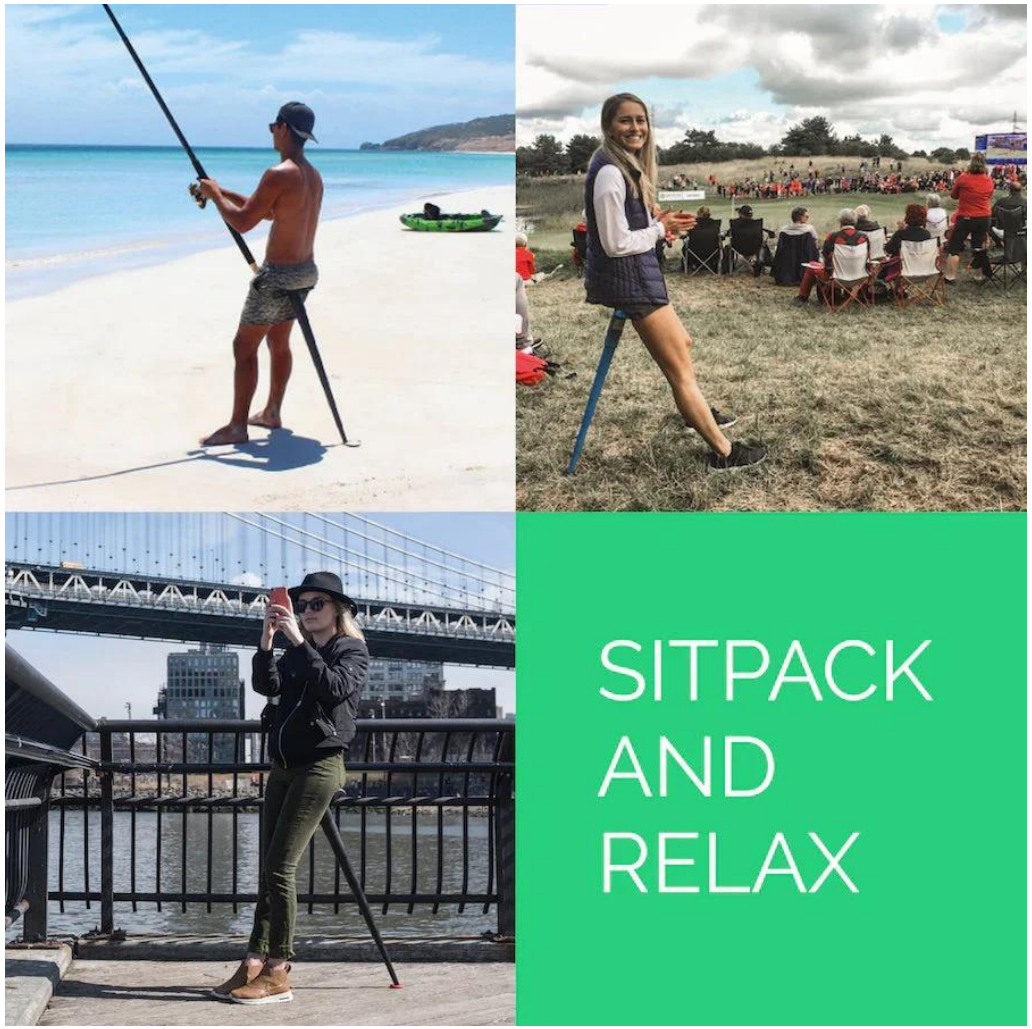




R – Respect Broader Impact

The Sitpack design reduces the need for bulkier, resource-heavy alternatives. It encourages outdoor activity by making seating available anywhere, supporting health and social benefits. Additionally, its durable design helps reduce waste by lasting longer than less durable options.



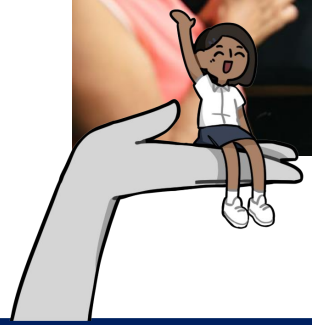


E - Enhance Lives

By providing a convenient and effective way for users to rest wherever they are, the Sitpack enhances lives by reducing fatigue and improving well-being. Its accessibility and portability ensure that users can maintain energy and comfort throughout their day.



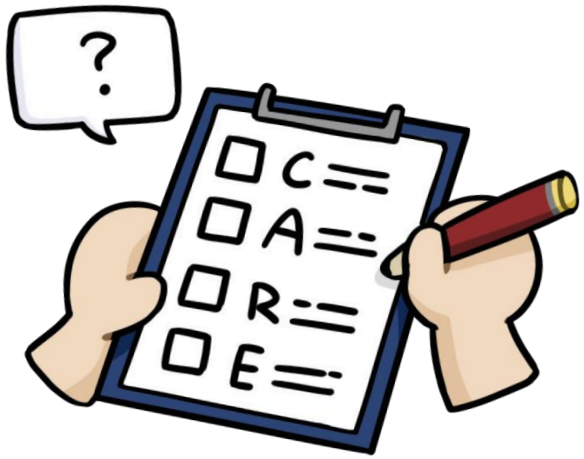




Design is everywhere around us!

Look around you and choose something you believe is a great design. Use the CARE framework to evaluate it:

- **C: Care for the user** – Does the design consider the user's needs, preferences, and experiences?
- **A: Achieve its goal** – Does the design solve the problem well and improve on what's already available?
- **R: Respect broader impact** – Does the design consider its broader effects on society, the environment, and long-term sustainability?
- **E: Enhance lives** – Does the design improve the user's quality of life, making it better, easier, or more enjoyable?





Explain why you think it is a good design, focusing on how it meets these criteria. Then, think about how it could be improved to make it even better!